Compton scattering 3

From the first section we have

$$\sum_{ab} \mathcal{M}_1{}^{\mu\nu} \mathcal{M}^*_{1\mu\nu} = \frac{e^4}{(s-m^2)^2} \left(-8su + 24m^2 s + 8m^2 u + 8m^4\right)$$

$$\sum_{ab} \mathcal{M}_1{}^{\mu\nu} \mathcal{M}^*_{2\nu\mu} = \frac{e^4}{(s-m^2)(u-m^2)} \left(8m^2 s + 8m^2 u + 16m^4\right)$$

$$\sum_{ab} \mathcal{M}_2{}^{\nu\mu} \mathcal{M}^*_{2\nu\mu} = \frac{e^4}{(u-m^2)^2} \left(-8su + 8m^2 s + 24m^2 u + 8m^4\right)$$

We also have by the Casimir trick

$$\sum_{ab} \mathcal{M}_1{}^{\mu\nu} \mathcal{M}^*_{1\mu\nu} = \frac{e^4}{(s-m^2)^2} \operatorname{Tr}\left((\not{q}_1 + m)\gamma^\mu(\not{p}_3 + m)\gamma_\mu(\not{q}_1 + m)\gamma^\nu(\not{p}_1 + m)\gamma_\nu\right)$$

$$\sum_{ab} \mathcal{M}_1{}^{\mu\nu} \mathcal{M}^*_{2\nu\mu} = \frac{e^4}{(s-m^2)(u-m^2)} \operatorname{Tr}\left((\not{q}_2 + m)\gamma^\mu(\not{p}_3 + m)\gamma^\nu(\not{q}_1 + m)\gamma_\mu(\not{p}_1 + m)\gamma_\nu\right)$$

$$\sum_{ab} \mathcal{M}_2{}^{\nu\mu} \mathcal{M}^*_{2\nu\mu} = \frac{e^4}{(u-m^2)^2} \left((\not{q}_2 + m)\gamma^\mu(\not{p}_3 + m)\gamma_\mu(\not{q}_2 + m)\gamma^\nu(\not{p}_1 + m)\gamma_\nu\right)$$

The trace operator Tr contracts (sums over) the first and last index. Slash matrices are formed as

$$\not{p} = p^\mu g_{\mu\nu} \gamma^\nu = p^0\gamma^0 - p^1\gamma^1 - p^2\gamma^2 - p^3\gamma^3$$

Eigenmath script